

1. DATASET FIELDS REFERENCED IN THIS GUIDE

Field names below match the NWIS drilling_timeseries schema exactly, so this guide can be cross-referenced directly against logged data.

Field Name	Unit	Typical Range	Description
Plastic_Viscosity	cP	18 - 40 cP	Resistance to flow from mechanical friction between mud particles
Yield_Point	lb/100ft2	14 - 26 lb/100ft2	Resistance to the start of flow; governs low-shear-rate carrying capacity
Mud_Weight	SG	1.05 - 1.44 SG	Static hydrostatic pressure gradient of the drilling fluid
Flow_Rate	gpm	550 - 700 gpm	Volume of mud pumped per unit time

2. HOW THESE FIELDS RELATE TO EACH OTHER

Plastic_Viscosity vs Standpipe_Pressure: Higher Plastic_Viscosity increases frictional pressure loss throughout the circulating system, raising Standpipe_Pressure and ECD for the same Flow_Rate - important in narrow pressure-margin sections where ECD must stay below the fracture gradient.

Yield_Point vs Cuttings Carrying Capacity: Yield_Point governs the mud's ability to suspend and carry cuttings at low annular velocities (e.g. during connections when Flow_Rate is briefly zero); a low Yield_Point relative to Inclination increases cuttings-bed formation risk even if Flow_Rate is otherwise adequate.

Plastic_Viscosity vs Yield_Point Ratio: The YP/PV ratio is commonly used as a quick indicator of a mud's low-shear-rate carrying efficiency; a declining YP/PV ratio over time (even with both individually in range) can signal degrading hole-cleaning performance before ROP or Torque symptoms appear.

Rheology vs Mud_Weight Changes: Diluting the mud system to reduce Mud_Weight (e.g. during a mud-loss response) typically reduces both Plastic_Viscosity and Yield_Point unless compensated with rheology-modifying additives, which can unintentionally weaken hole cleaning at the same time losses are being managed.

3. KEY FORMULAS / RULES OF THUMB

Formula / Rule	Notes
YP/PV Ratio = Yield_Point / Plastic_Viscosity	A ratio trending down over time at similar Flow_Rate/Inclination suggests declining hole-cleaning efficiency.
Approximate Frictional Pressure Loss increases with both Flow_Rate and Plastic_Viscosity	Used qualitatively: raising either Flow_Rate or Plastic_Viscosity to fix hole cleaning will also raise Standpipe_Pressure/ECD.
Minimum recommended Yield_Point (rule of thumb) scales with Inclination	Higher-inclination intervals require a higher Yield_Point target to keep the same margin against cuttings-bed formation.

4. RISK INDICATORS & RECOMMENDED ACTIONS

Observed Pattern (Fields Involved)	Risk Indicated	Recommended Action
Plastic_Viscosity and Yield_Point both dropping after a Mud_Weight reduction	Reduced hole-cleaning capacity following a loss-control dilution	Add rheology-restoring products; do not rely on Flow_Rate alone to compensate

Standpipe_Pressure rising at constant Flow_Rate after a rheology treatment	Frictional pressure loss increased with Plastic_Viscosity - ECD may be approaching fracture gradient	Re-check ECD/PWD reading against the fracture gradient before continuing at the same Flow_Rate
YP/PV ratio declining over several days at similar Inclination	Gradually degrading cuttings-carrying capacity	Schedule a mud treatment (Yield_Point-focused) before a hole-cleaning-related Torque/Hook_Load issue appears
Low Yield_Point relative to a high-Inclination interval	Elevated cuttings-bed / pack-off risk	Increase Yield_Point via treatment and/or increase hi-vis sweep frequency

5. SAMPLE LOG ROWS (drilling_timeseries SCHEMA)

Well_ID	Timestamp	Depth_MD	Mud_Weight	Plastic_Viscosity	Yield_Point
W001	2024-05-15 06:00	2905	1.18	24	18
W002	2024-10-02 02:00	3160	1.32	32	22

6. FIELD_ALPHA CROSS-REFERENCE

- W001 (Alpha-01): Mud_Weight was reduced from 1.22 to 1.18 SG at Depth_MD 2905 m to control losses; Plastic_Viscosity and Yield_Point were held at 24 cP / 18 lb/100ft2 through the treatment rather than allowed to drop further, preserving hole-cleaning capacity.
- W002 (Alpha-02): Yield_Point was increased from 20 to 22-23 lb/100ft2 around the stuck-pipe event at Depth_MD 3160 m as part of the hole-cleaning response in the 28-deg tangent section.

7. OPERATIONAL BEST PRACTICES

- Track the YP/PV ratio as a derived field, not just the two raw values, to catch early hole-cleaning degradation.
- When reducing Mud_Weight for loss control, treat rheology maintenance as a parallel task rather than an afterthought.
- Set Yield_Point targets relative to Inclination, not a single fixed field-wide number.

This Drilling Parameter Reference Guide is a synthetically generated reference document produced for the NWIS prototype, aligned to the drilling_timeseries dataset schema. Not associated with an actual drilling operation.